Handout 3: Experimental Study

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1 Student Outcomes

By the end of this activity, students will be able to:

1. Develop an experimental hypothesis based on prior literature and scientific reasoning.
2. Design a structured experimental protocol including measurements, analysis methods, and safety considerations.
3. Critically evaluate the feasibility, limitations, and ethical considerations of a proposed experimental study.

2 Equipment

1. Use Google Scholar to download papers related to your project topic. https://scholar.google.com
2. Accessing research papers.
   (a) Students working on campus and connected to UIC WiFi can usually download papers directly through the university library subscriptions.
   (b) Students working off campus can access papers using one of the following methods:
      i. Use the UIC VPN service to connect to the university network. Instructions are available here:
      https://it.uic.edu/services/faculty-staff/uic-network/uic-vpn/
      ii. Use the UIC Library EZProxy service. Instructions are available here:
      https://library.uic.edu/help-articles/proxy-urls-ezproxy-2/

3 Experimental Protocol

Based on the papers you reviewed, you will now develop your own experimental protocol. You should propose a hypothesis and design an experiment that could test this hypothesis. Use the same structured approach that you used during the extensive literature review. Keep your notes concise, organized, and scientifically clear. Note that this is an initial draft of your experimental protocol and it will change as you do experiments.

1. **Hypothesis**

   State the main hypothesis or research question for your proposed study.

   What are you trying to test or demonstrate?

2. **Experiment**

   Describe the experiment or study that you plan to perform.

   Under what conditions will the experiment be conducted?

   Examples:

   - Walking on a treadmill
   - Walking at different speeds
   - Comparing different exercise conditions
   - Measuring physiological response during activity

3. **Measurements**

   What variables or quantities will be measured?

   Examples:

   - Speed
   - Heart rate
   - Step count
   - Muscle activity

   What devices or sensors will be used?

   Examples:

   - Microcontroller
   - Heart rate sensor
   - IMU sensor
   - Motion capture system

   Also identify where the sensors or devices will be mounted.

4. **Analysis**

   Describe how the collected data will be analyzed.

   Identify:

   - Metrics that will be computed
   - Number and type of subjects
   - Statistical tools or methods

   Examples:

- Average heart rate over a 2-minute interval
- Comparison between conditions
- Student t-test
- ANOVA

5. **Safety and Ethics**

   Identify possible safety concerns or risks in your study.

   Would this study require IRB approval if conducted on human subjects?

   Describe any precautions that should be taken.

6. **Expected Results**

   What results do you expect to observe if your hypothesis is correct?

   What measurements or analyses would support your conclusions?

7. **Limitations**

   What limitations or weaknesses might exist in your proposed study?

   What factors could affect the quality or reliability of the results?

Show your work to the teaching assistant.